



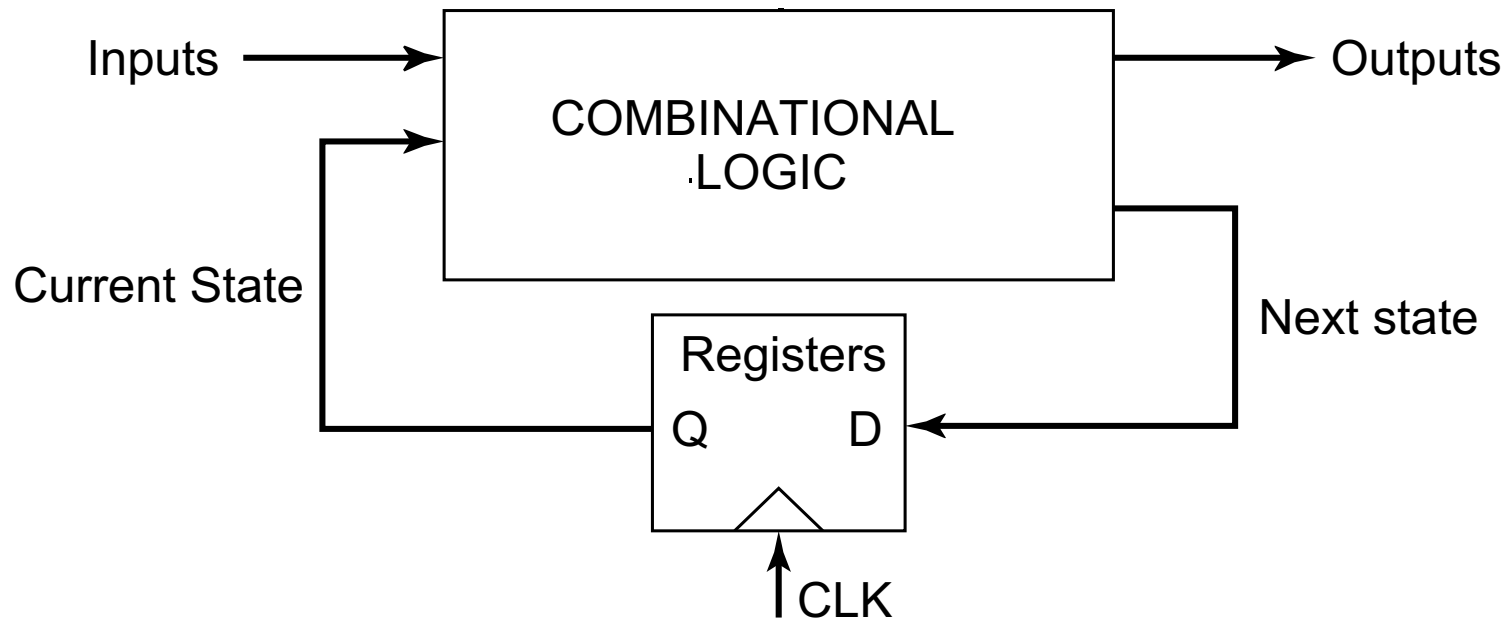
Digital Integrated Circuits

A Design Perspective

Jan M. Rabaey
Anantha Chandrakasan
Borivoje Nikolic

Designing Sequential Logic Circuits

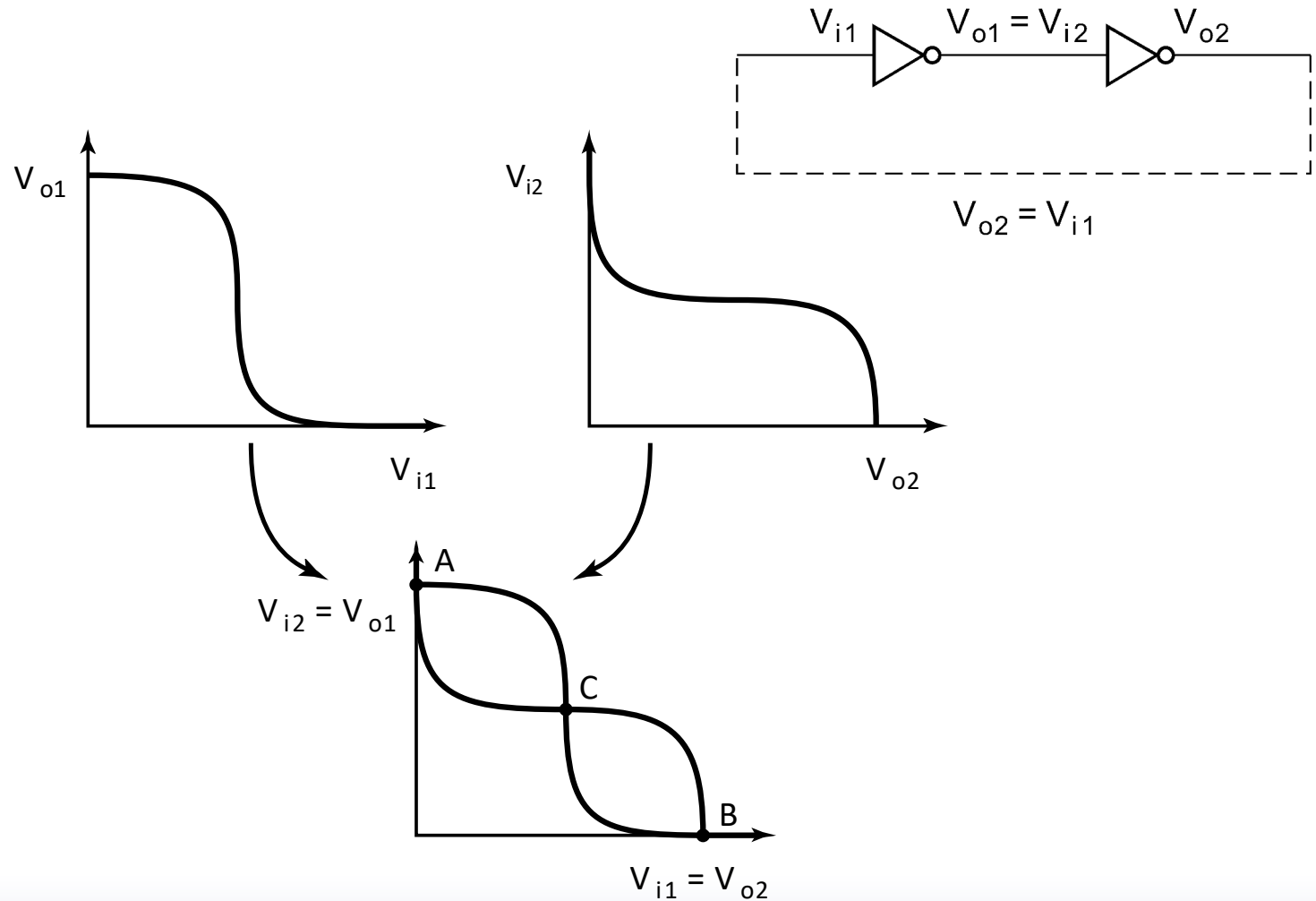
Sequential Logic



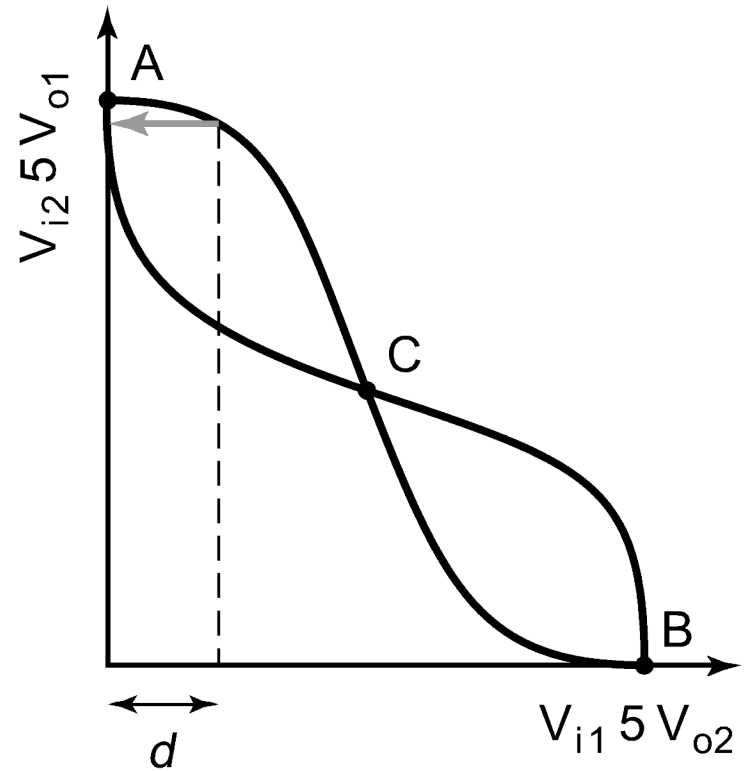
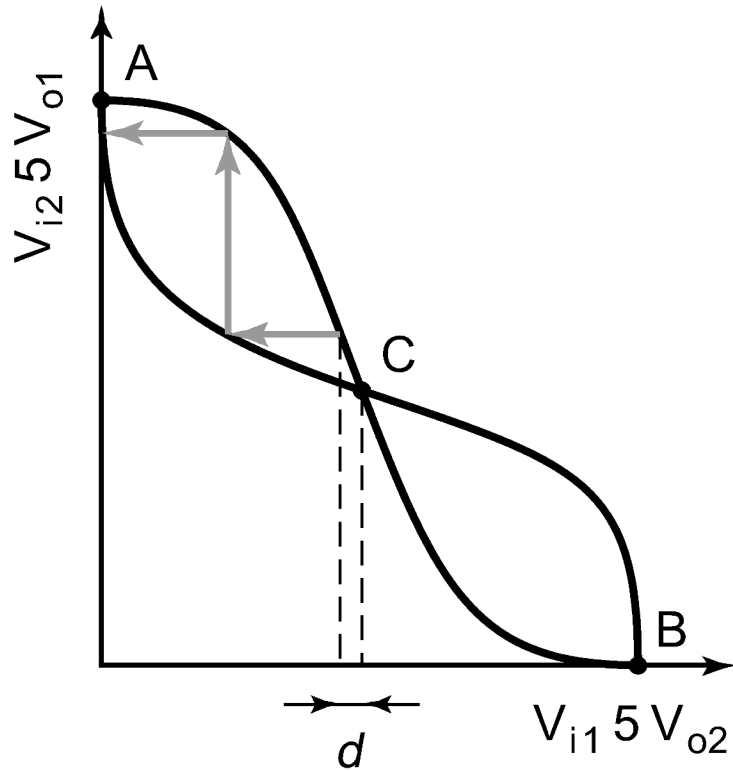
2 storage mechanisms

- positive feedback
- charge-based

Positive Feedback: Bi-Stability



Meta-Stability



Gain should be larger than 1 in the transition region

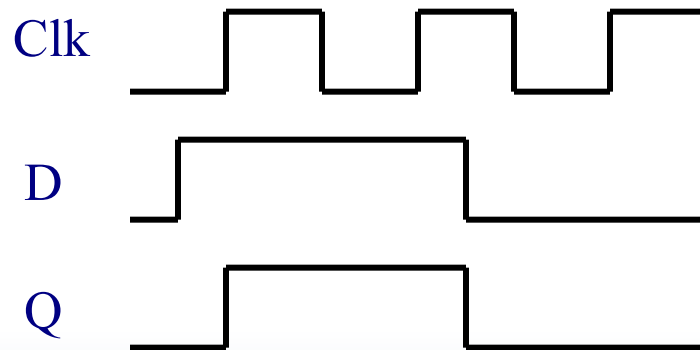
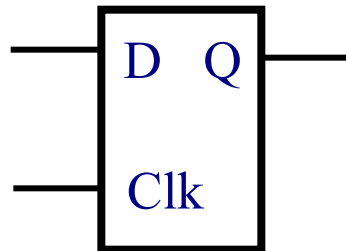
Naming Conventions

- In our text:
 - a latch is level sensitive
 - a register is edge-triggered
- There are many different naming conventions
 - For instance, many books call edge-triggered elements flip-flops
 - This leads to confusion however

Latch versus Register

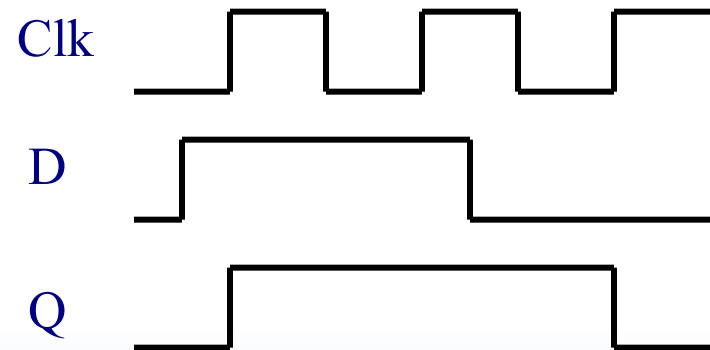
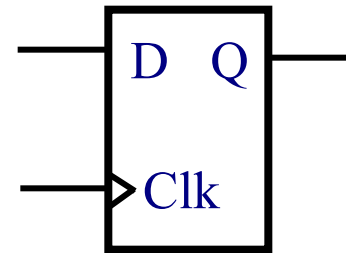
❑ Latch

stores data when
clock is low



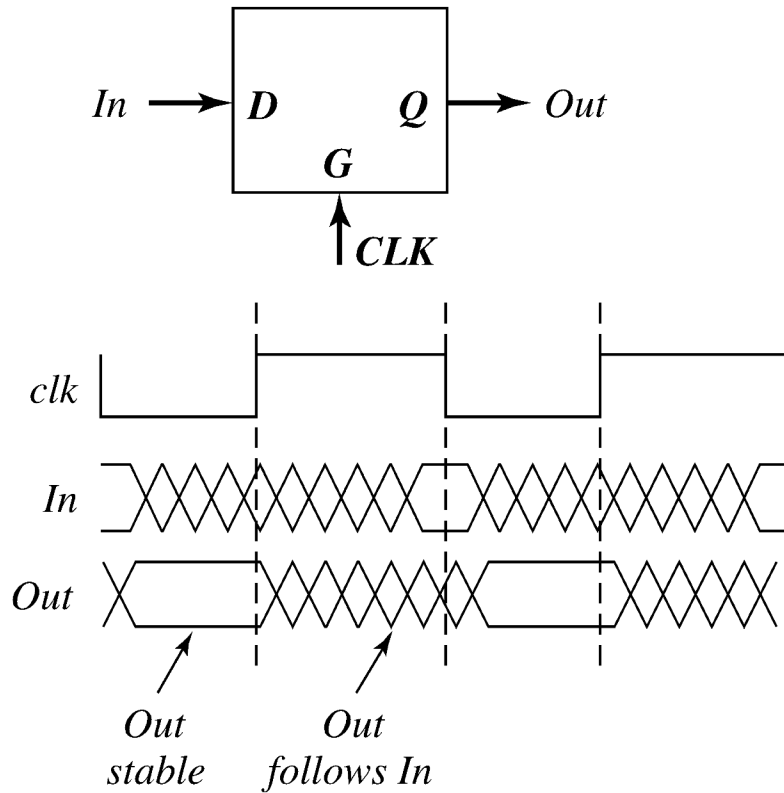
❑ Register

stores data when
clock rises

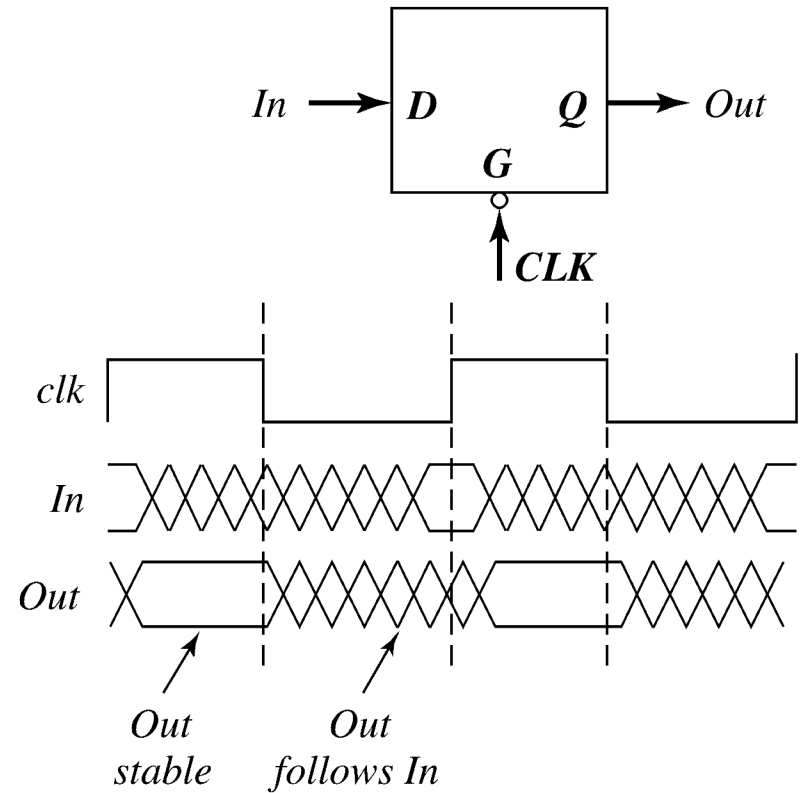


Latches

Positive Latch



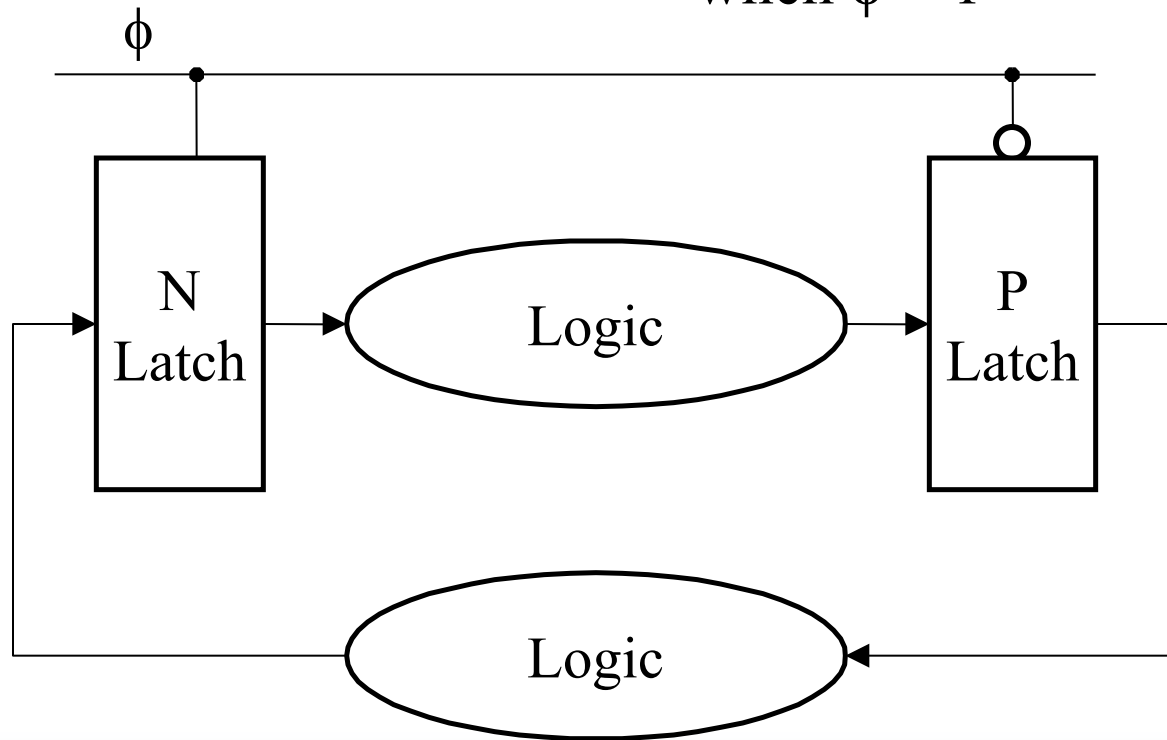
Negative Latch



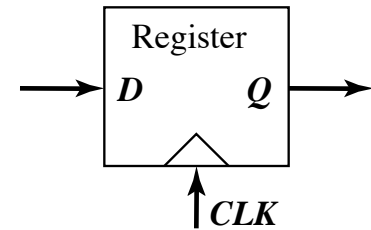
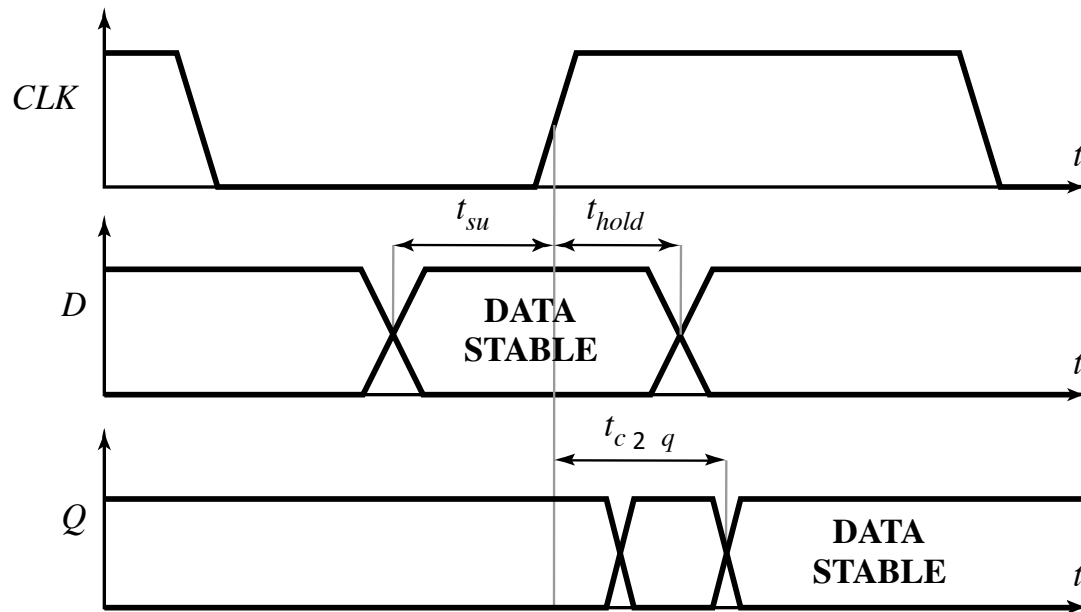
Latch-Based Design

- N latch is transparent when $\phi = 0$

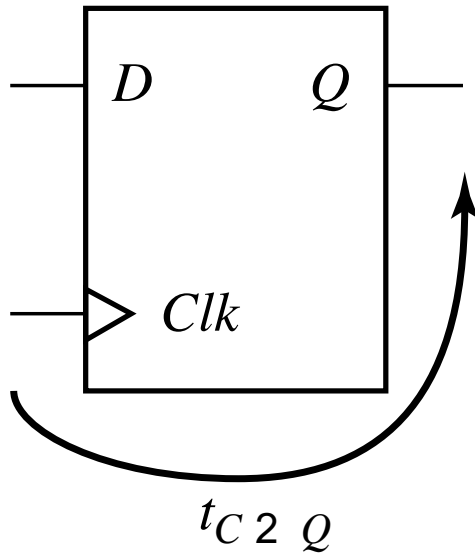
- P latch is transparent when $\phi = 1$



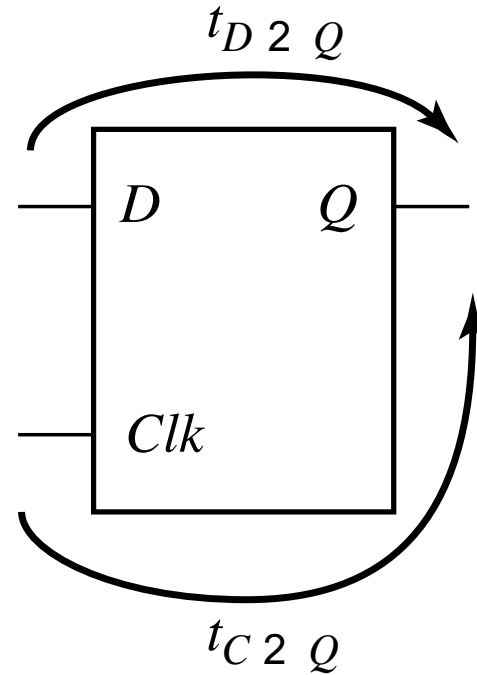
Timing Definitions



Characterizing Timing

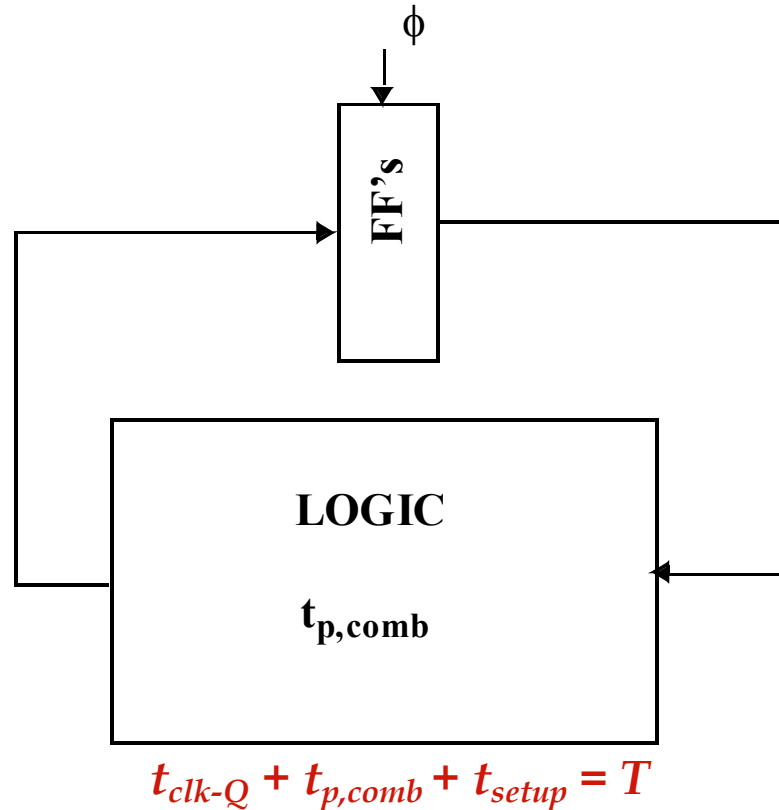


Register

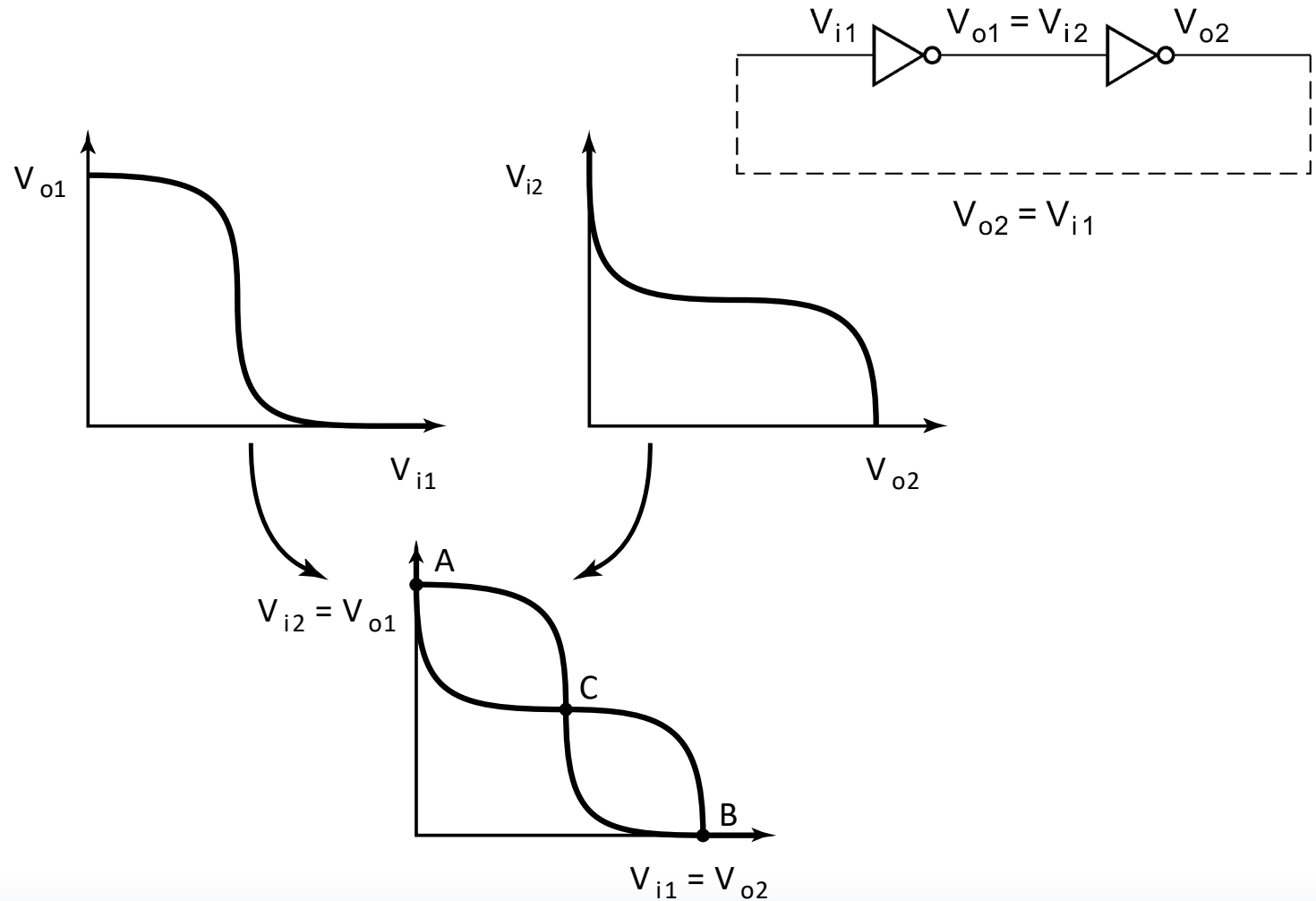


Latch

Maximum Clock Frequency

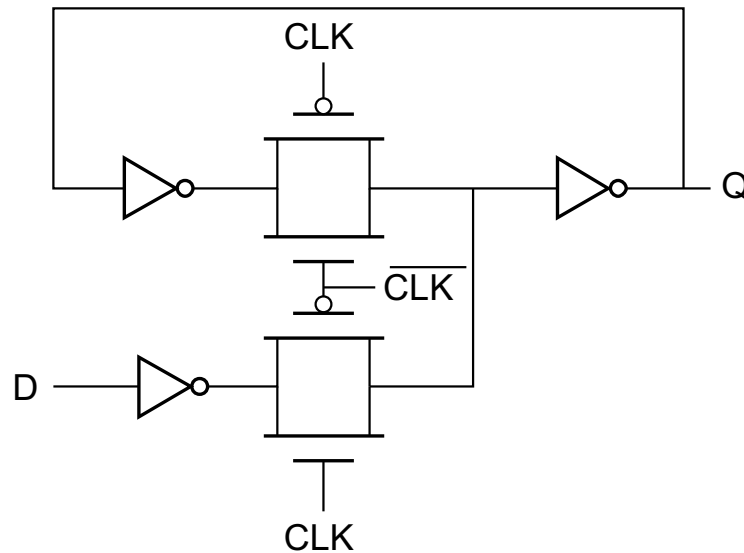


Positive Feedback: Bi-Stability

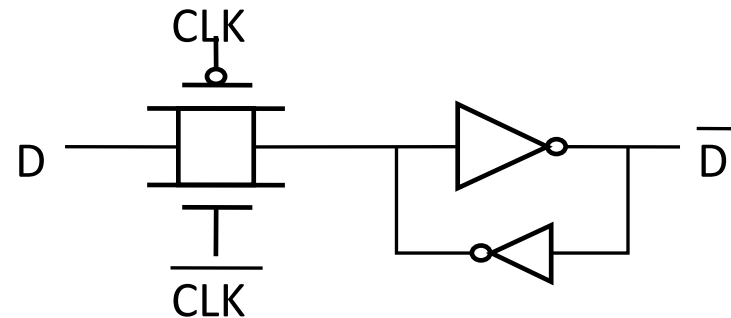


Writing into a Static Latch

Use the clock as a decoupling signal,
that distinguishes between the transparent and opaque states



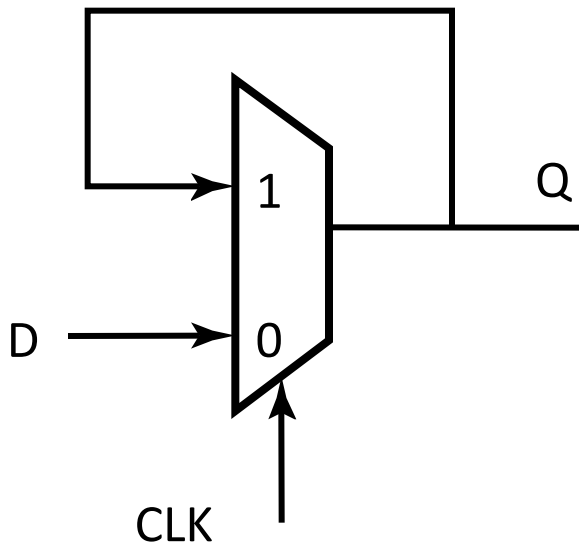
Converting into a MUX



Forcing the state
(can implement as NMOS-only)

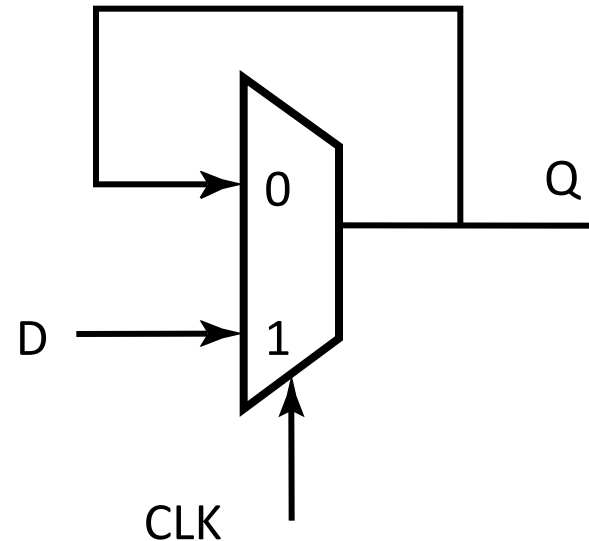
Mux-Based Latches

Negative latch
(transparent when CLK= 0)



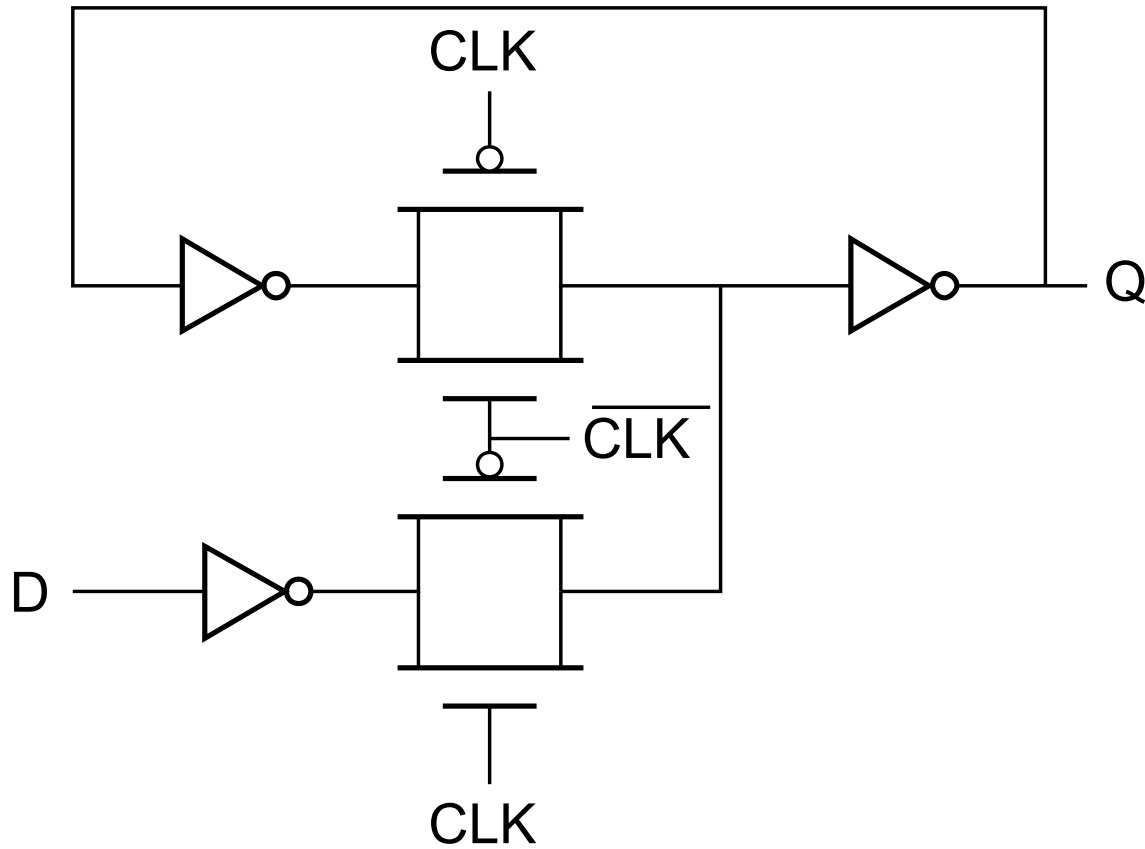
$$Q = \overline{Clk} \cdot Q + Clk \cdot In$$

Positive latch
(transparent when CLK= 1)

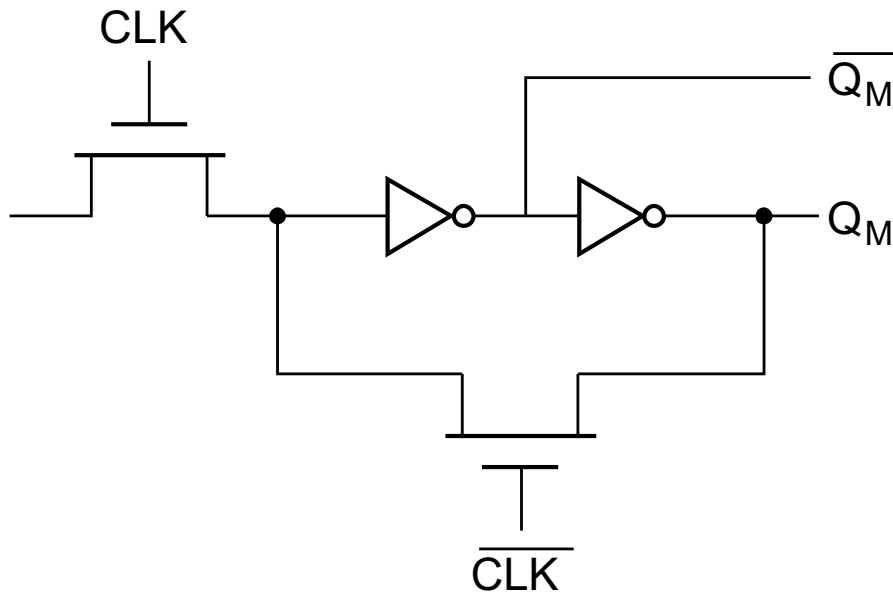


$$Q = Clk \cdot Q + \overline{Clk} \cdot In$$

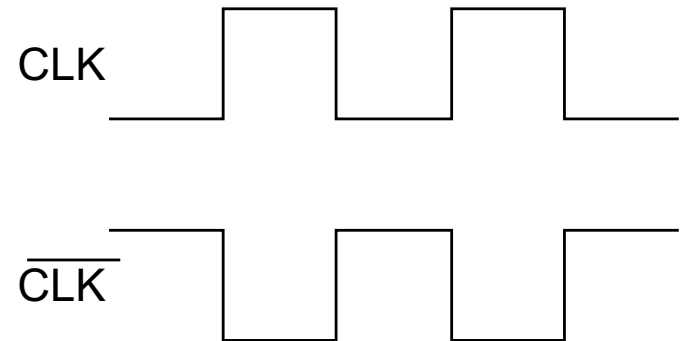
Mux-Based Latch



Mux-Based Latch (pseudo-static)

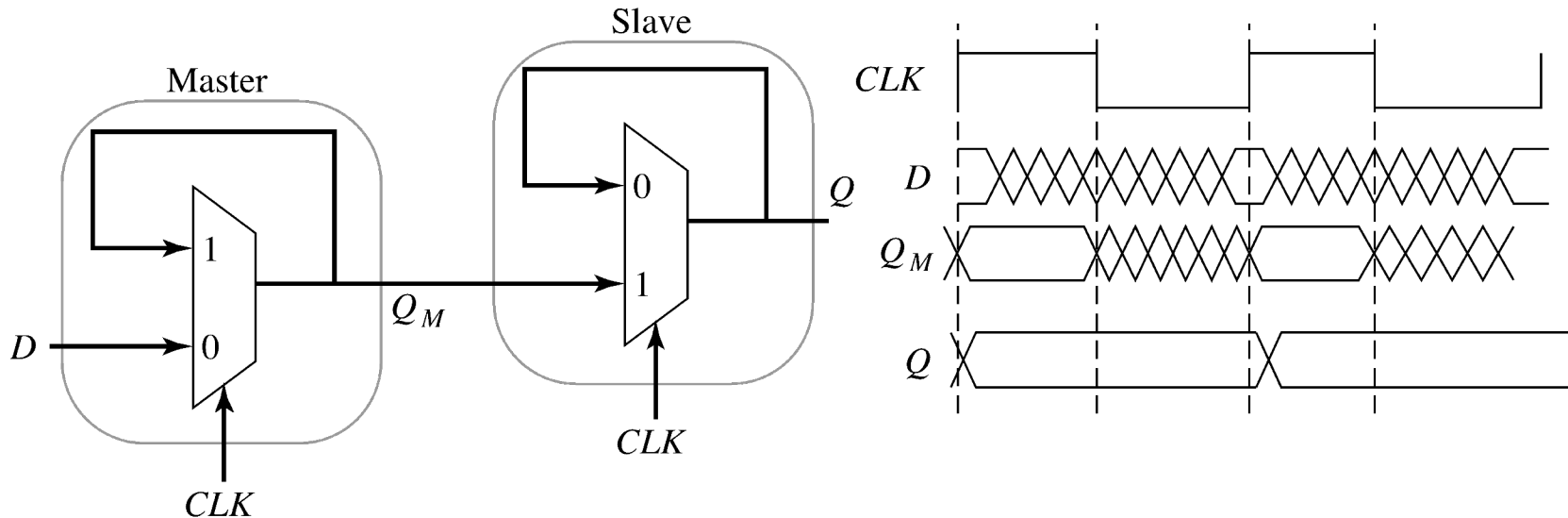


NMOS only



Non-overlapping clocks

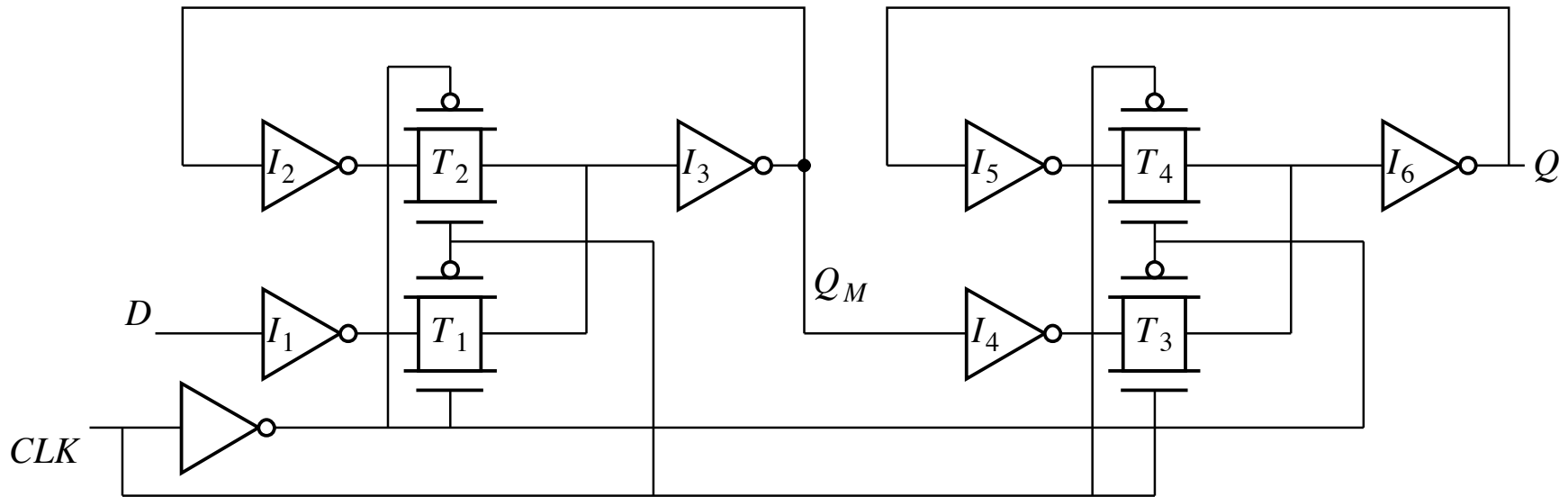
Master-Slave (Edge-Triggered) Register



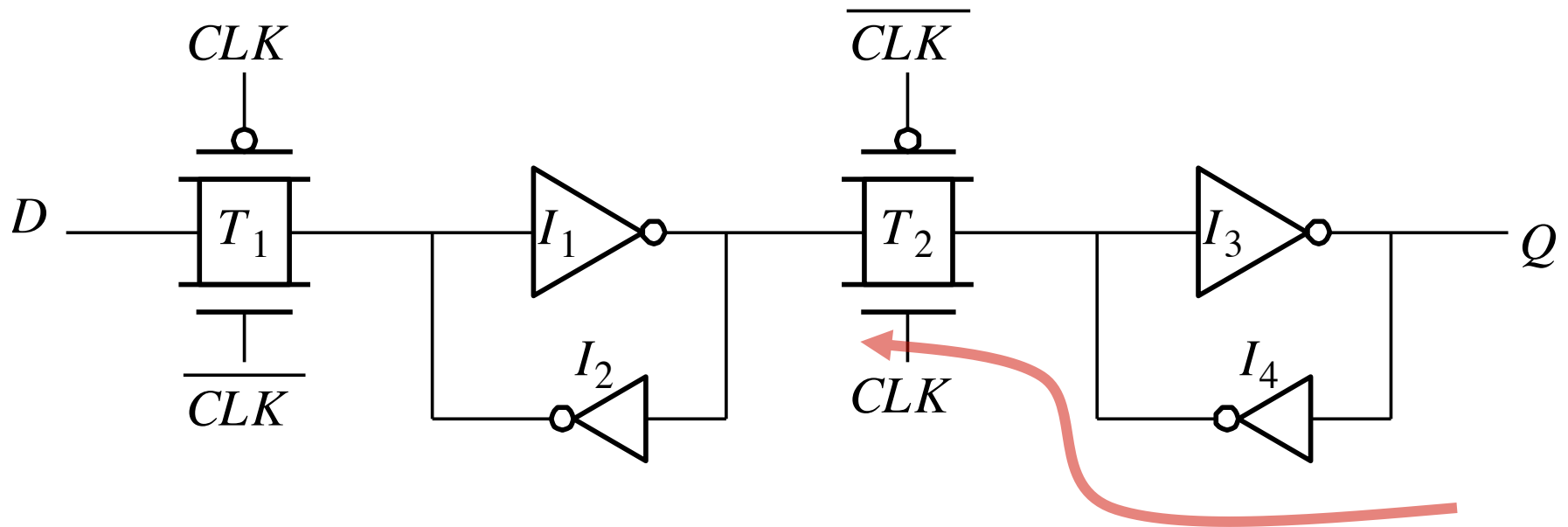
Two opposite latches trigger on edge
Also called master-slave latch pair

Master-Slave Register

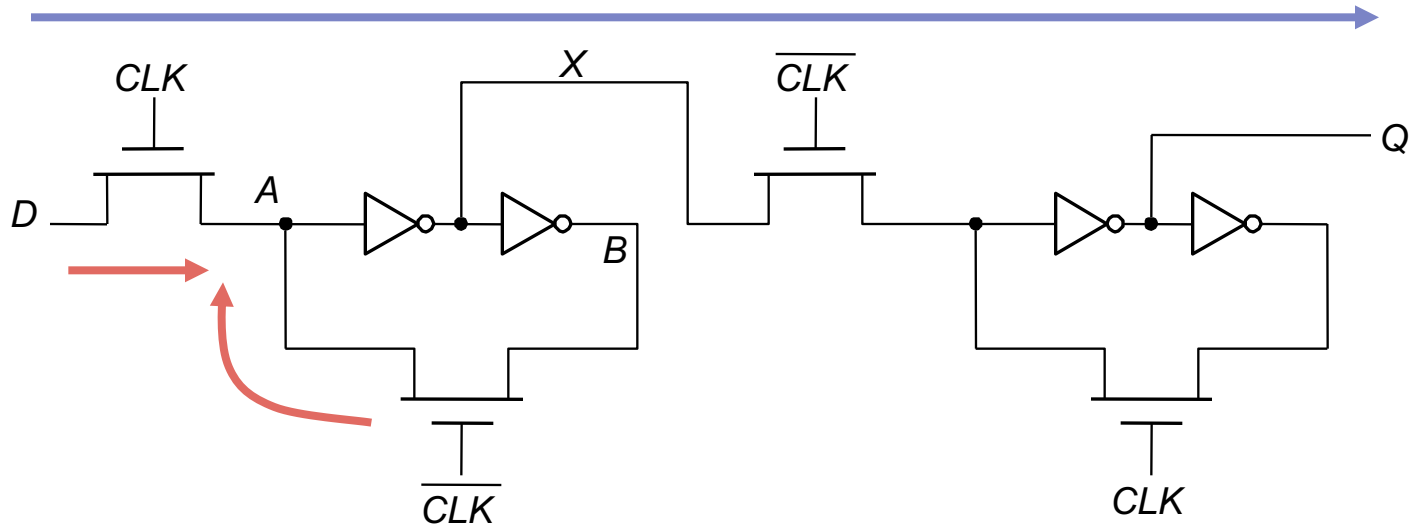
Multiplexer-based latch pair



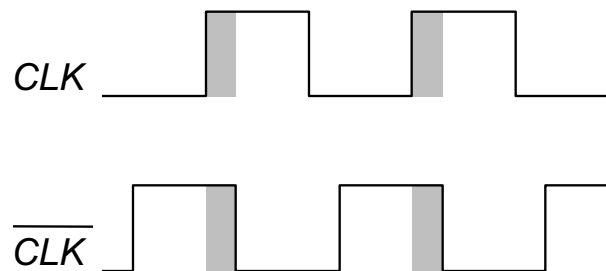
Reduced Clock Load Master-Slave Register



Avoiding Clock Overlap



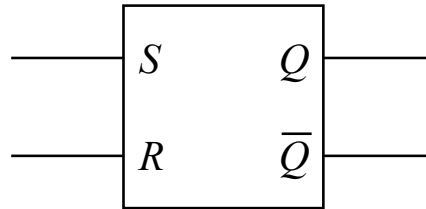
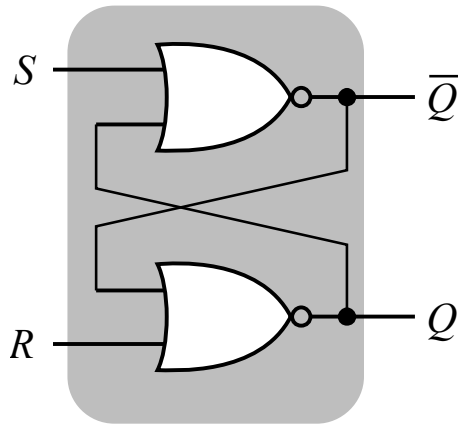
(a) Schematic diagram



(b) Overlapping clock pairs

Overpowering the Feedback Loop — Cross-Coupled Pairs

NOR-based set-reset

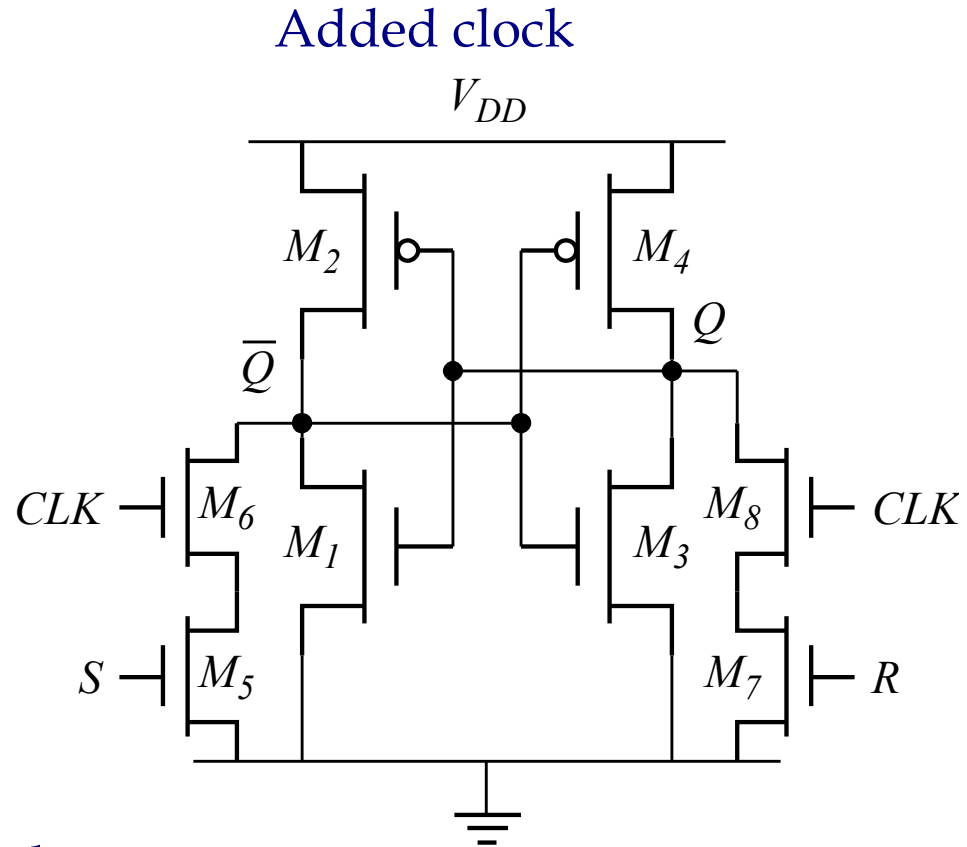
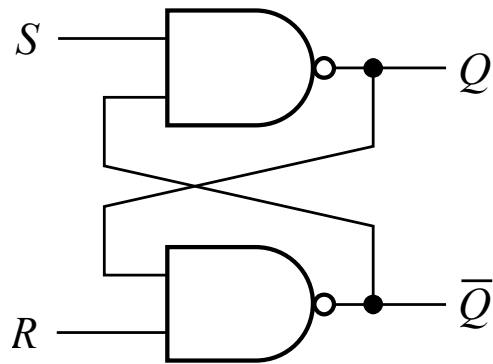


S	R	Q	$\bar{Q}$
0	0	Q	$\bar{Q}$
1	0	1	0
0	1	0	1
1	1	0	0

Forbidden State

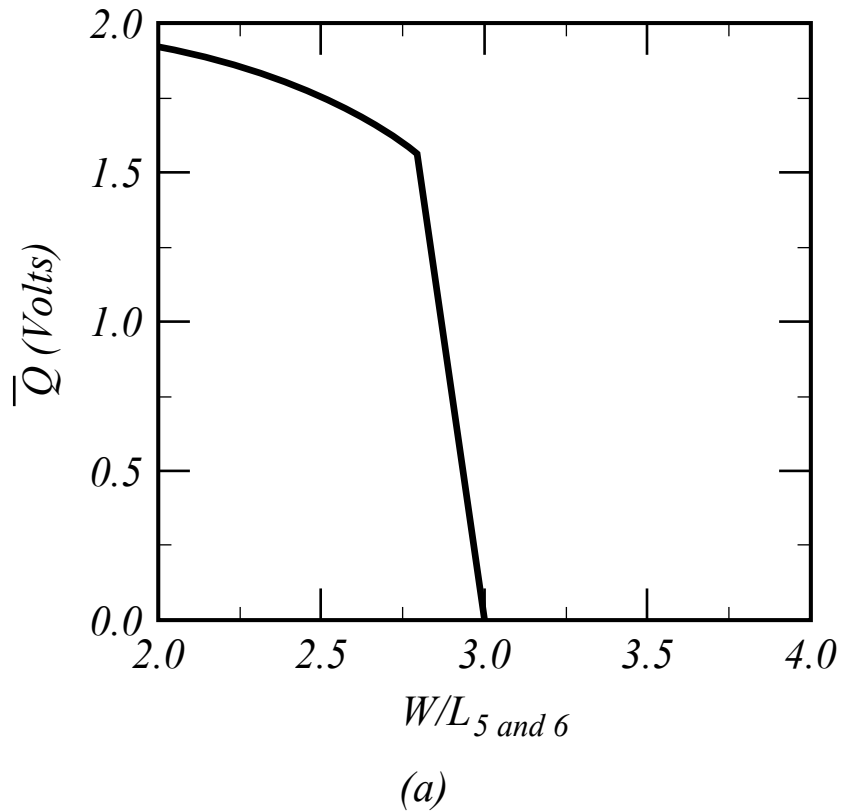
Cross-Coupled NAND

Cross-coupled NANDs

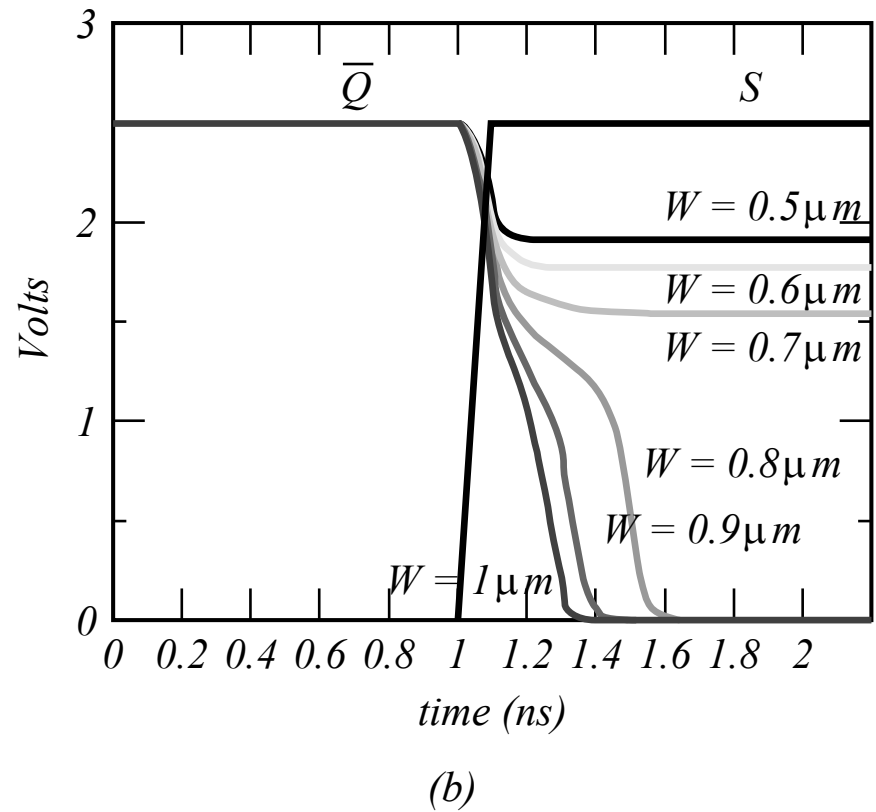


This is not used in datapaths anymore,
but is a basic building memory cell

Sizing Issues

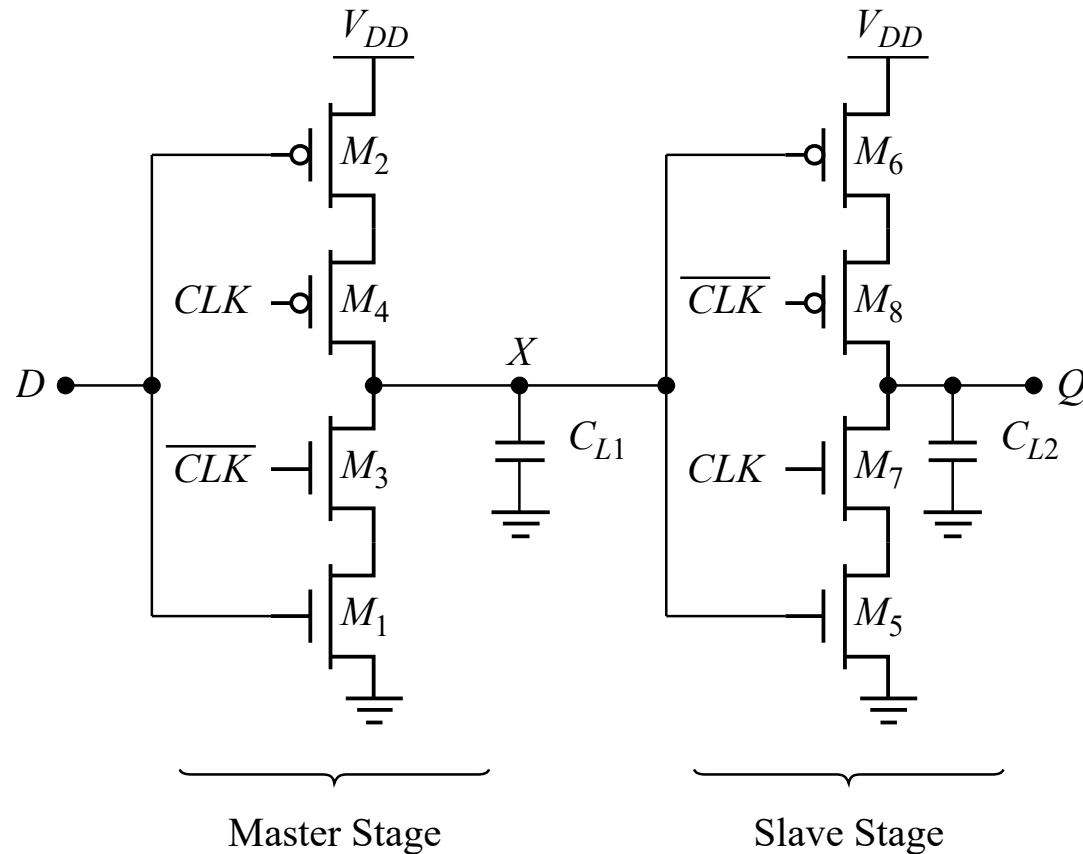


Output voltage dependence
on transistor width

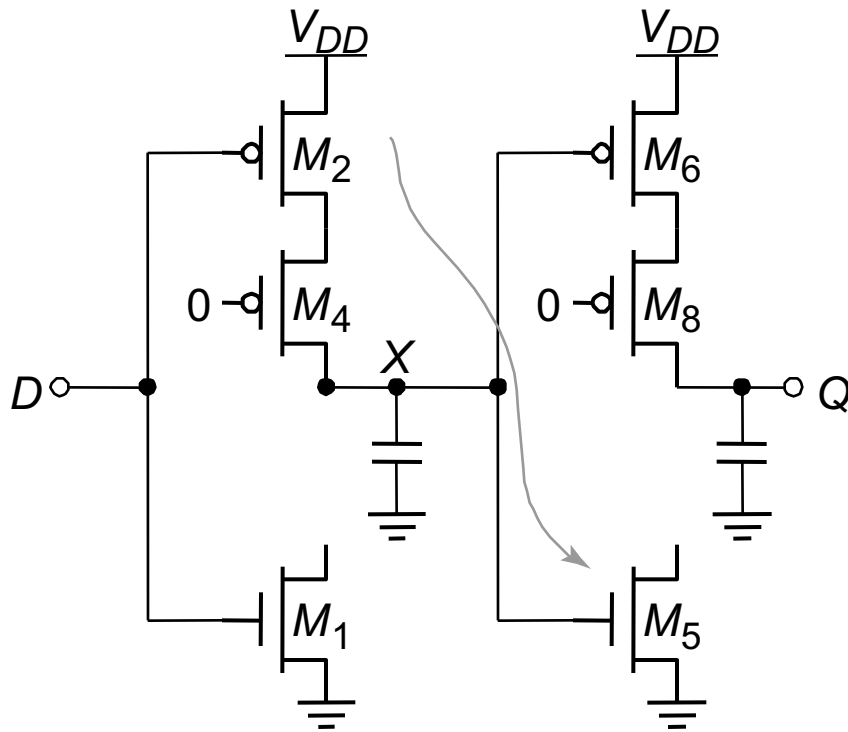


Transient response

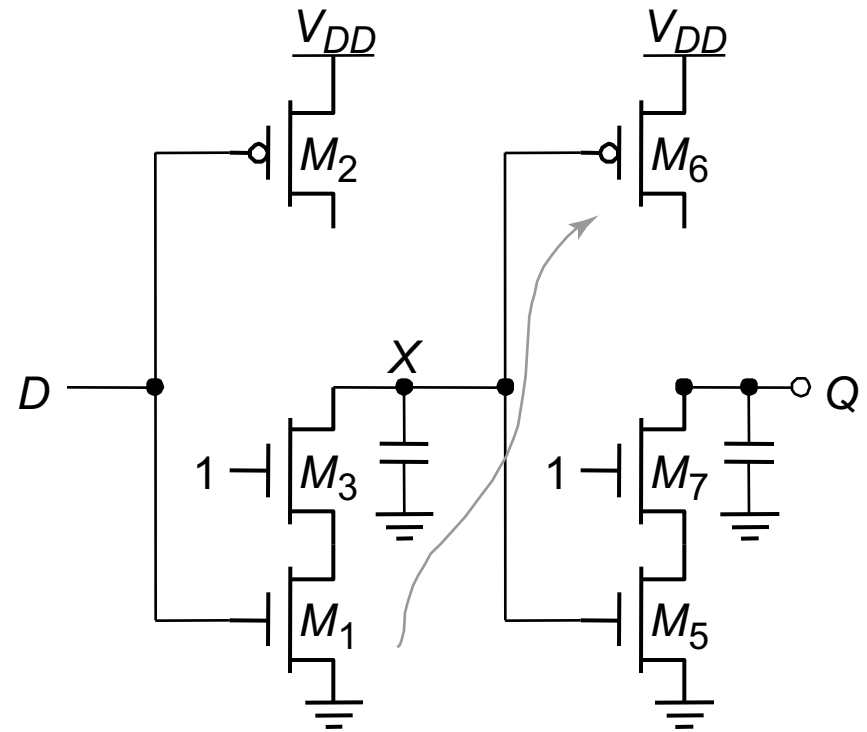
Other Latches/Registers: C²MOS



Insensitive to Clock-Overlap

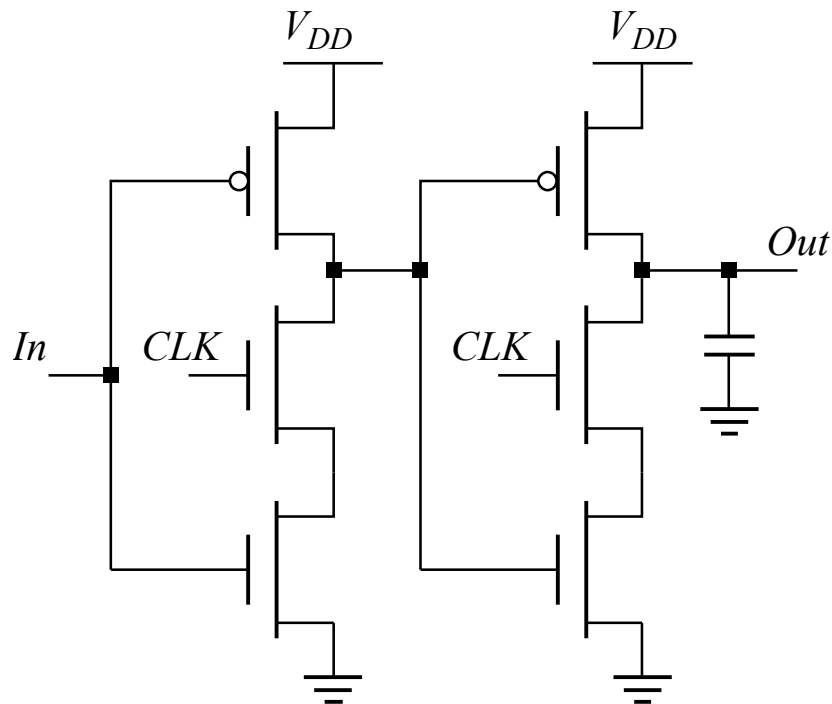


(a) (0-0) overlap



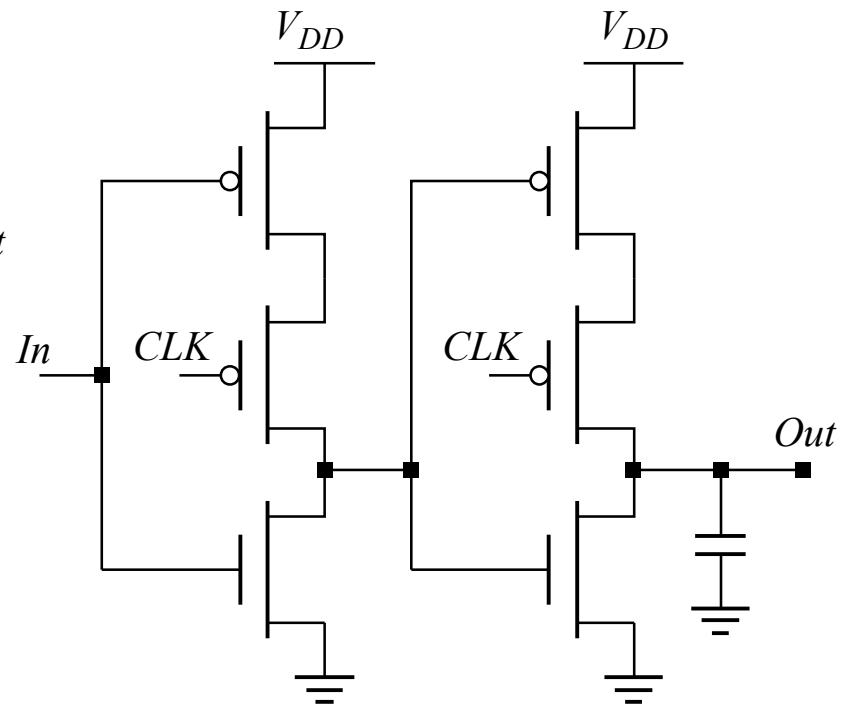
(b) (1-1) overlap

Other Latches/Registers: TSPC



Positive latch

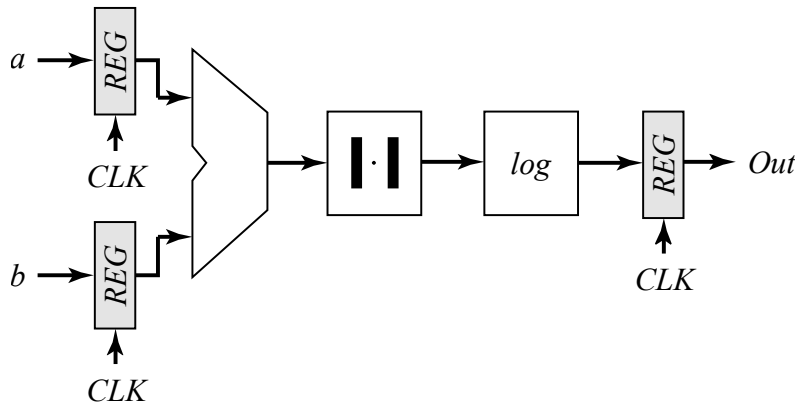
(transparent when $CLK = 1$)



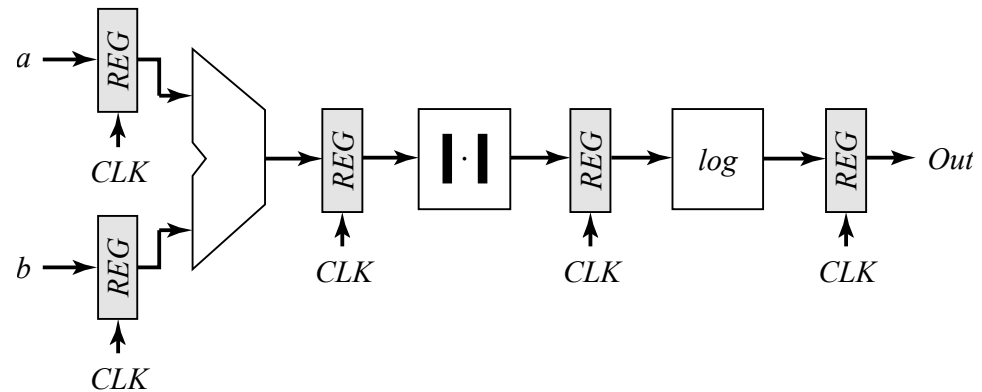
Negative latch

(transparent when $CLK = 0$)

Pipelining



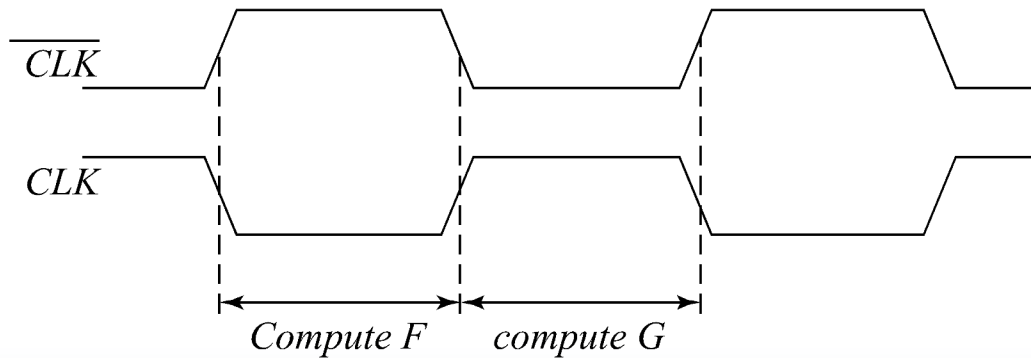
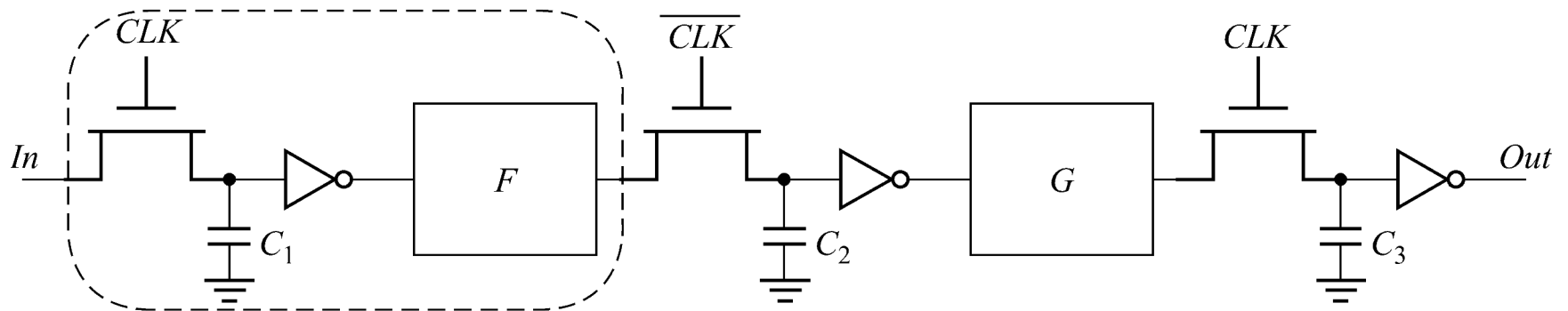
Reference



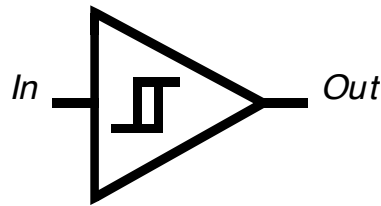
Pipelined

Clock Period	Adder	Absolute Value	Logarithm
1	$a_1 + b_1$		
2	$a_2 + b_2$	$ a_1 + b_1 $	
3	$a_3 + b_3$	$ a_2 + b_2 $	$\log(a_1 + b_1)$
4	$a_4 + b_4$	$ a_3 + b_3 $	$\log(a_2 + b_2)$
5	$a_5 + b_5$	$ a_4 + b_4 $	$\log(a_3 + b_3)$

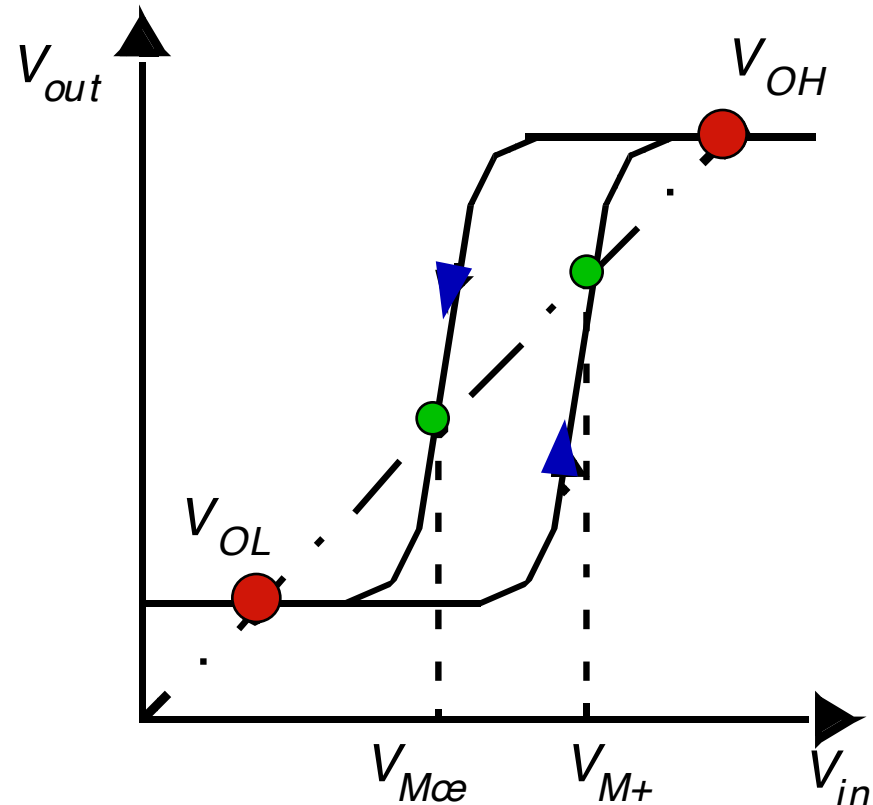
Latch-Based Pipeline



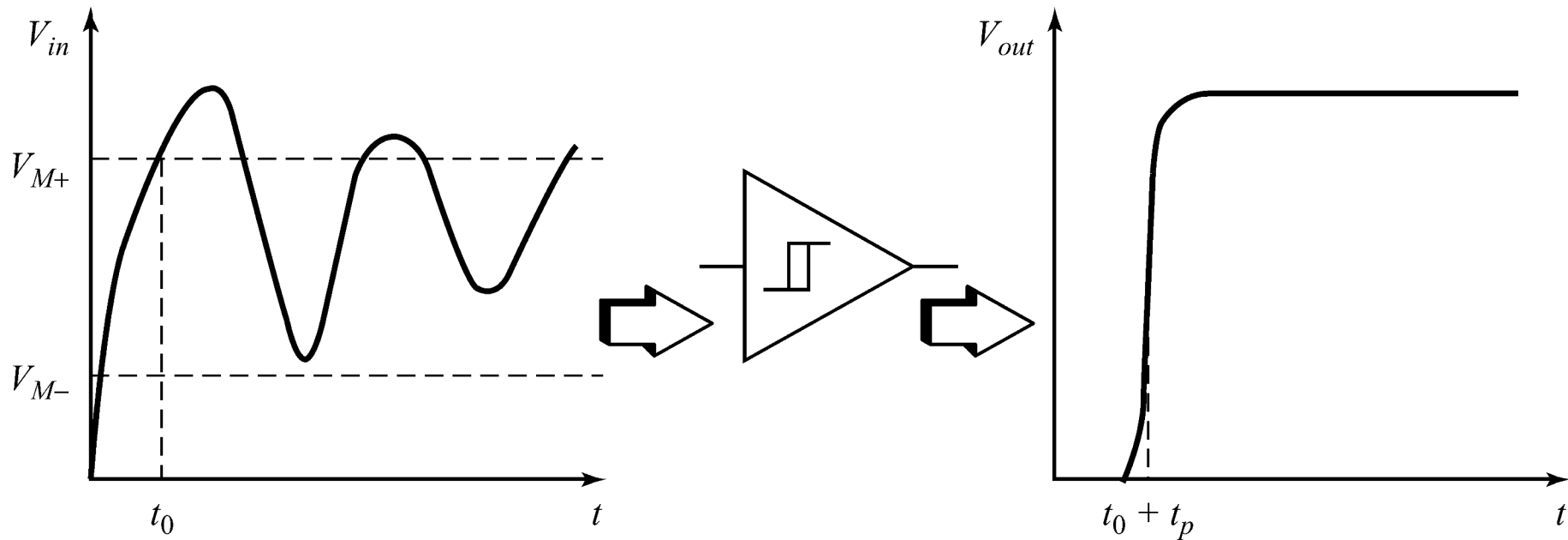
Non-Bistable Sequential Circuits— Schmitt Trigger



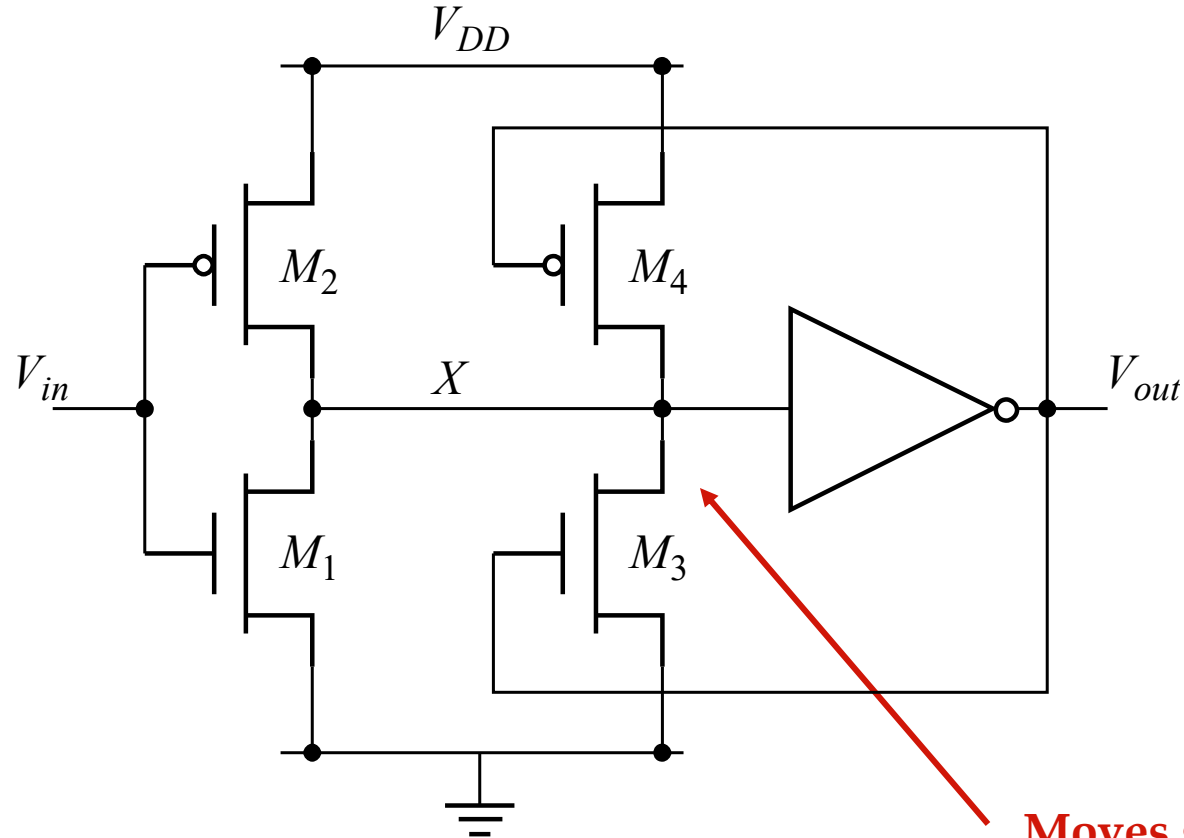
- VTC with hysteresis
- Restores signal slopes



Noise Suppression using Schmitt Trigger

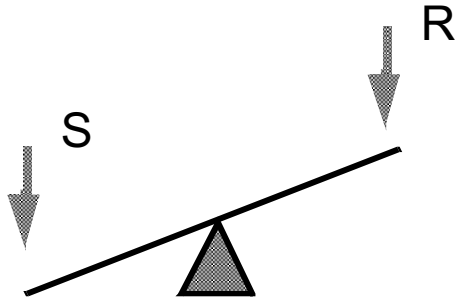


CMOS Schmitt Trigger

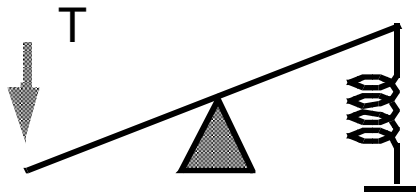


**Moves switching threshold
of the first inverter**

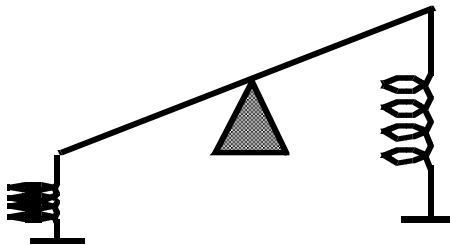
Multivibrator Circuits



Bistable Multivibrator
flip-flop, Schmitt Trigger

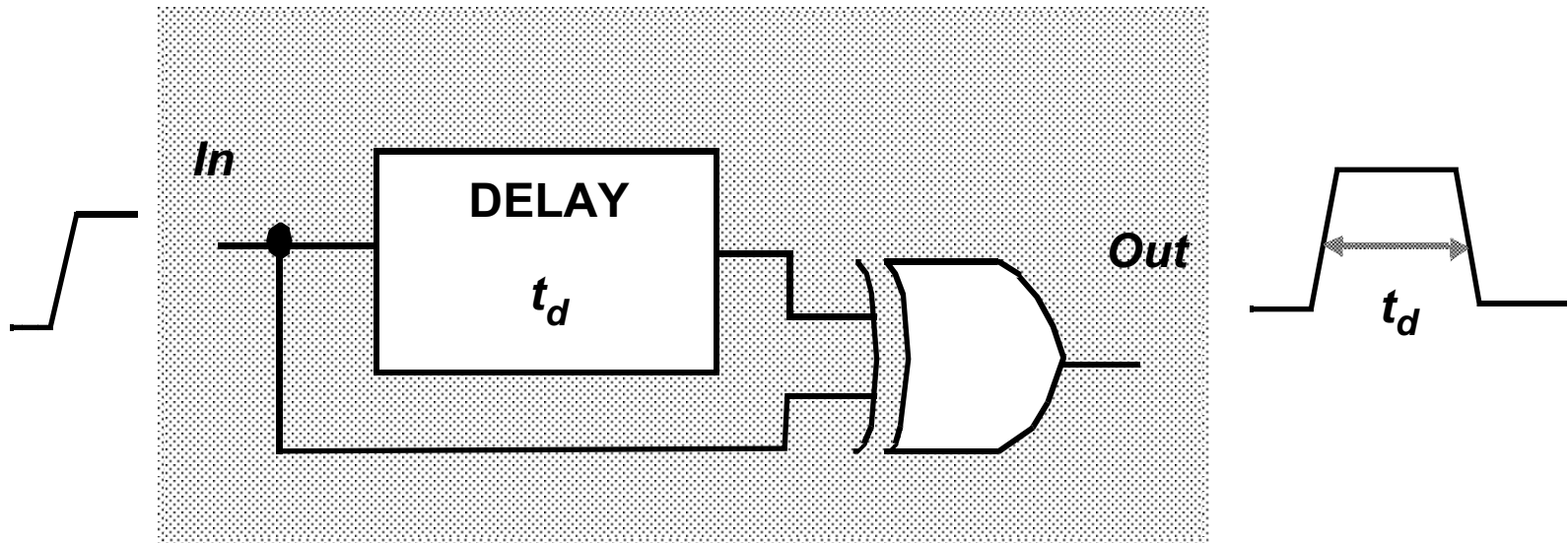


Monostable Multivibrator
one-shot

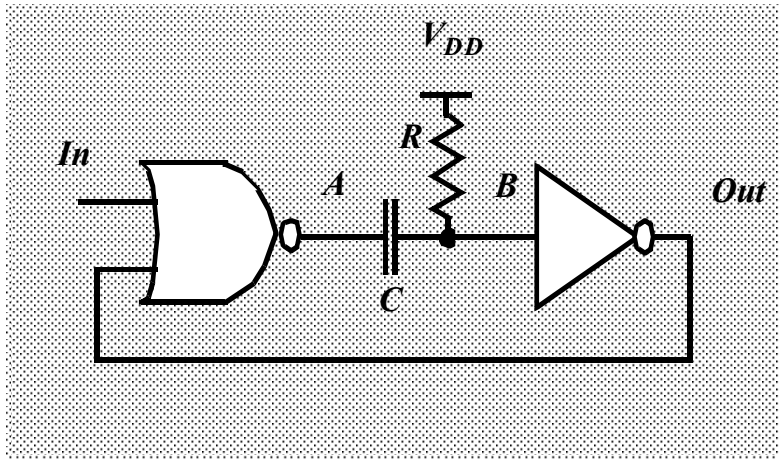


Astable Multivibrator
oscillator

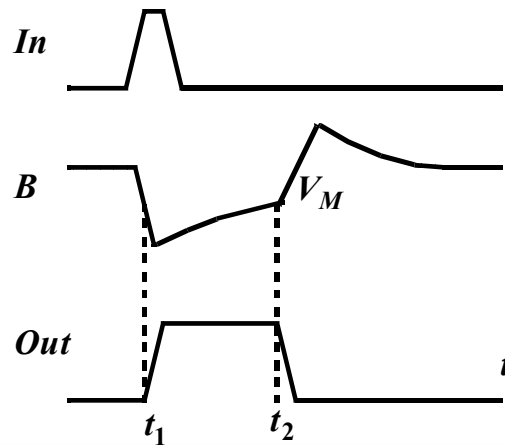
Transition-Triggered Monostable



Monostable Trigger (RC-based)

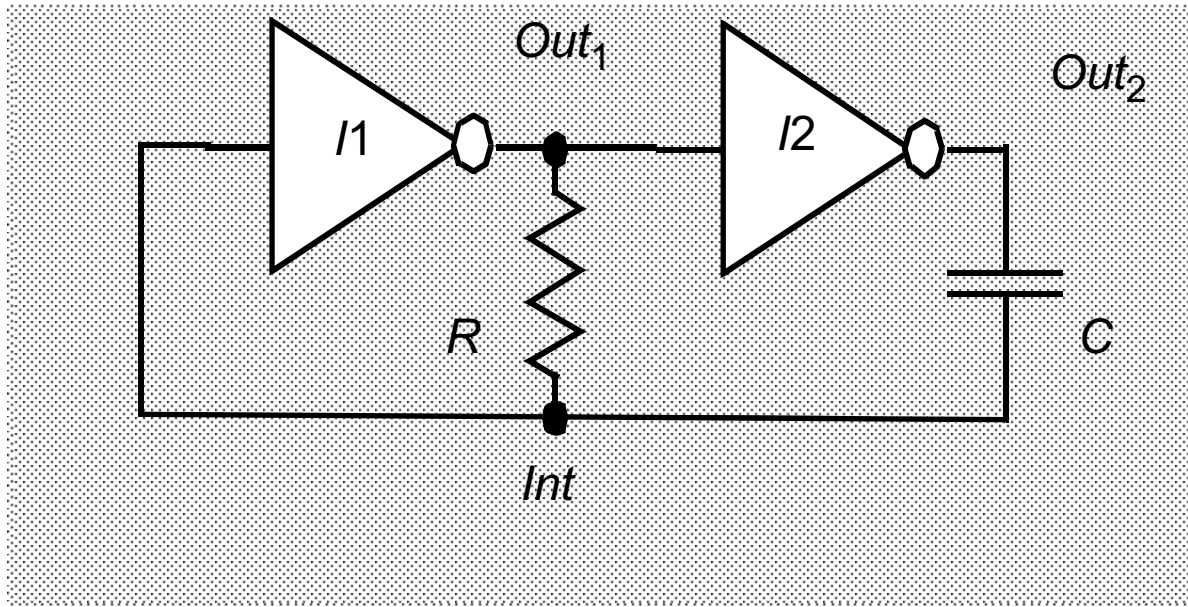


(a) Trigger circuit.



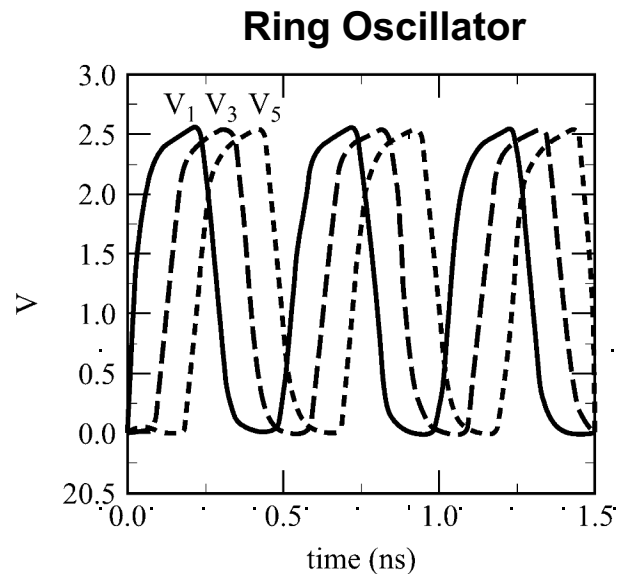
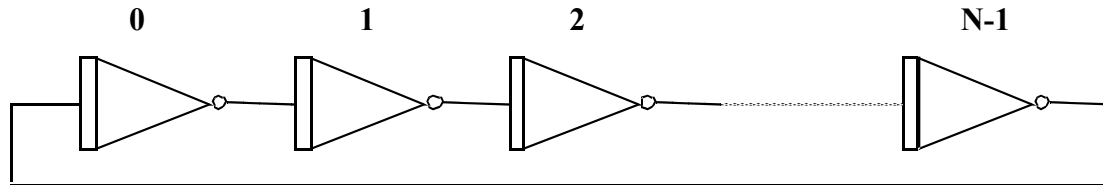
(b) Waveforms.

Relaxation Oscillator



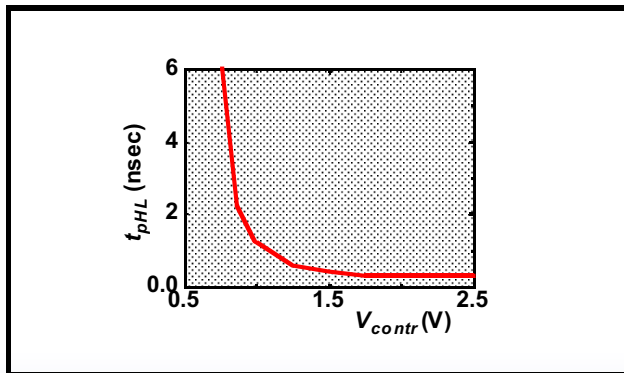
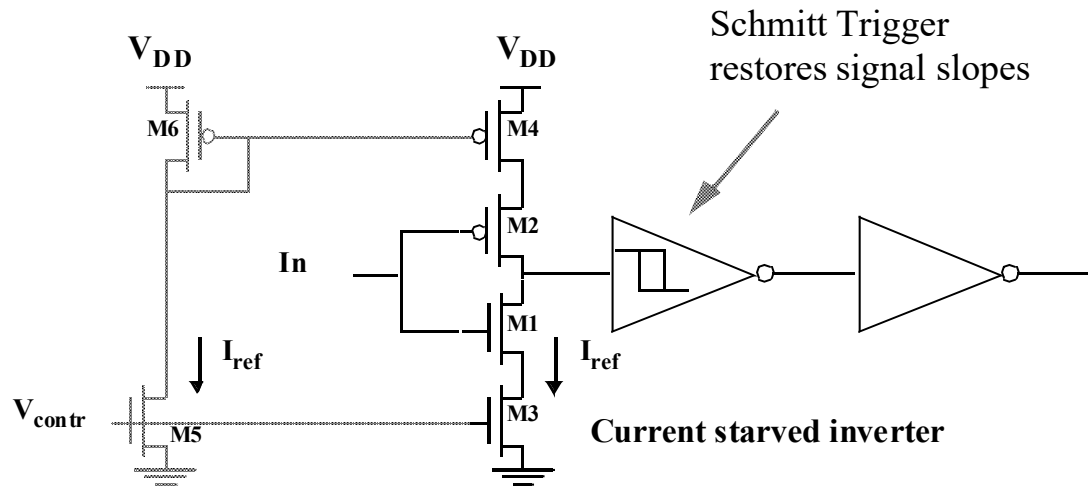
$$T = 2 (\log 3) RC$$

Astable Multivibrators (Oscillators)



simulated response of 5-stage oscillator

Voltage Controller Oscillator (VCO)



propagation delay as a function of control voltage